

Fundamental limits and asymptotic estimates



Outline

Fundamental limits

Asymptotic estimates

Asymptotic estimates and graphs

Fundamental limits

Introduction: Fundamental limits

- ▶ We now look at some common techniques for computing limits that combine the general theorems on limits with the use of some **fundamental limits** of certain elementary functions
- ▶ We start from the fundamental limits of **sine** and **cosine**, which are indeterminate forms $\left[\frac{0}{0}\right]$
- ▶ From the fundamental limit defining the number e we will then deduce the fundamental limits of the **logarithm**, the **exponential** and the **power**
- ▶ These limits are the basis of the **asymptotic estimates** that we will use to compute limits and sketch graphs

Fundamental limits of sine and cosine

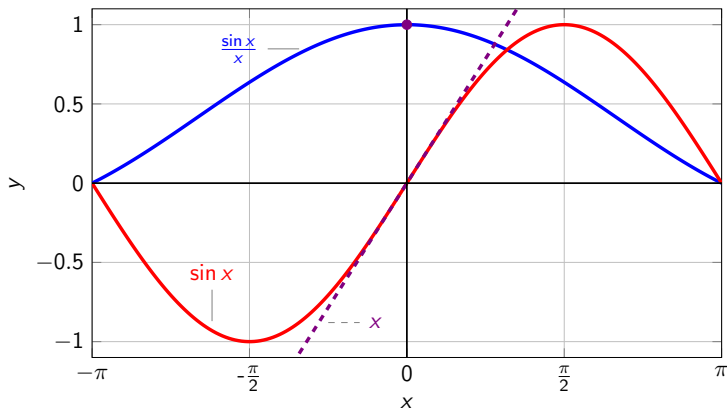
- We want to compute

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = \left[\frac{0}{0} \right] \quad \text{and} \quad \lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} = \left[\frac{0}{0} \right]$$

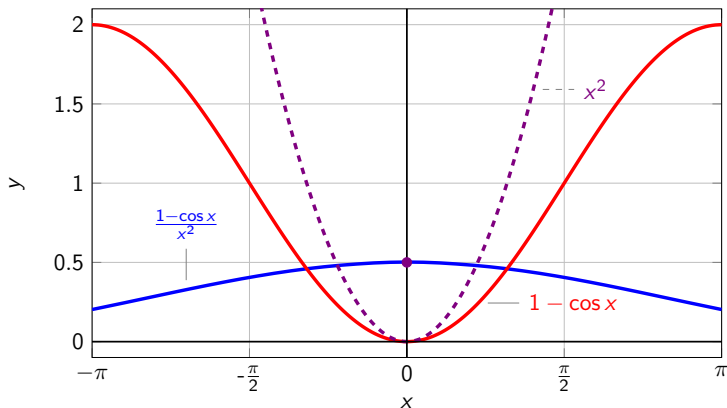
which are indeterminate forms

- Before proving the value of these limits, let us look at the graphs of the functions involved

Graph of $\frac{\sin x}{x}$



Graph of $\frac{1-\cos x}{x^2}$



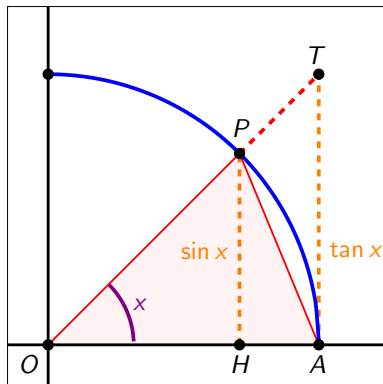
First fundamental limit

Lemma 1

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

Proof (1/3)

The functions $\sin x$ and x are odd functions, hence $\frac{\sin x}{x}$ is an even function. Therefore it is sufficient to compute $\lim_{x \rightarrow 0^+} \frac{\sin x}{x}$. We have:



$$HP = \sin x \quad AT = \tan x \quad \widehat{AP} = x$$

Proof (2/3)

The area of the triangle OPA is smaller than that of the circular sector OPA , which in turn is smaller than that of the triangle OTA . It follows that

$$\underbrace{\frac{1}{2} \cdot 1 \cdot \sin x}_{\text{area triang. } OPA} \leq \underbrace{\frac{1}{2} \cdot 1 \cdot x}_{\text{area circ. sector } OPA} \leq \underbrace{\frac{1}{2} \cdot 1 \cdot \tan x}_{\text{area triang. } OTA}$$

that is, for $x \in (0, \frac{\pi}{2})$:

$$\sin x < x < \tan x$$

Proof (3/3)

Dividing by $\sin x$, which is positive because $x \in (0, \frac{\pi}{2})$, we have

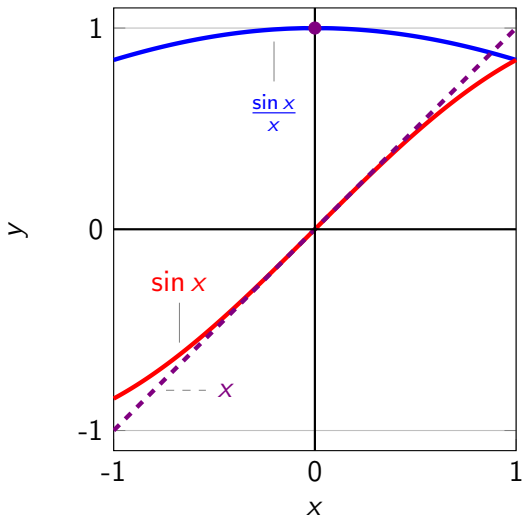
$$1 < \frac{x}{\sin x} < \frac{1}{\cos x}, \quad \forall x \in \left(0, \frac{\pi}{2}\right)$$

that is

$$\cos x < \frac{\sin x}{x} < 1, \quad \forall x \in \left(0, \frac{\pi}{2}\right)$$

By the comparison theorem, since $\lim_{x \rightarrow 0} \cos x = 1$, the lemma follows. □

Graph of $\frac{\sin x}{x}$ near 0



Second fundamental limit

Lemma 2

$$\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} = \frac{1}{2}$$

Proof

We consider

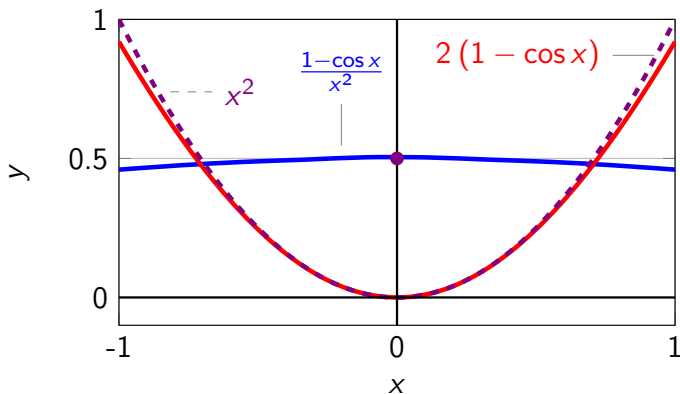
$$\frac{1 - \cos x}{x^2} = \frac{1 - \cos^2 x}{x^2 (1 + \cos x)} = \left(\frac{\sin x}{x} \right)^2 \frac{1}{1 + \cos x}$$

hence, since

$$\frac{\sin x}{x} \rightarrow 1 \quad \text{and} \quad (1 + \cos x) \rightarrow 2 \quad \text{as} \quad x \rightarrow 0,$$

the claim follows. □

Graph of $\frac{1-\cos x}{x^2}$ near 0



Continuous extension of a function

- ▶ Based on the fundamental limits just proved, the functions $f(x) = \frac{\sin x}{x}$ and $g(x) = \frac{1-\cos x}{x^2}$, initially not defined at $x = 0$, can be extended by continuity also at $x = 0$, by setting

$$f(x) = \begin{cases} \frac{\sin x}{x} & \text{if } x \neq 0 \\ 1 & \text{if } x = 0 \end{cases} \quad g(x) = \begin{cases} \frac{1-\cos x}{x^2} & \text{if } x \neq 0 \\ \frac{1}{2} & \text{if } x = 0 \end{cases}$$

- ▶ The functions f and g defined in this way are continuous also at $x = 0$

Extension by continuity

- ▶ If a function $f(x)$ is not defined at x_0 but the finite limit exists

$$\lim_{x \rightarrow x_0} f(x) = \ell$$

the function can be extended by continuity also at x_0 , by defining

$$f(x_0) = \ell$$

- ▶ If instead the function f has at x_0 a jump discontinuity, a vertical asymptote, or in any case does not have a finite limit, it is not possible to make it continuous at x_0 by changing its definition at a single point

Other fundamental limits

- ▶ We know that for every sequence $\{a_n\}$ diverging to $+\infty$ or $-\infty$ we have

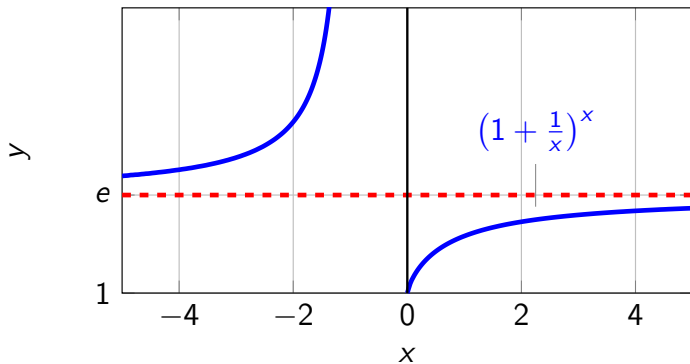
$$\lim_{n \rightarrow +\infty} \left(1 + \frac{1}{a_n}\right)^{a_n} = e$$

- ▶ By the sequential definition of the limit of a function, this fact immediately implies the next fundamental limit

Lemma 3

$$\lim_{x \rightarrow \pm\infty} \left(1 + \frac{1}{x}\right)^x = e$$

Graph of $(1 + \frac{1}{x})^x$



Generalization

- ▶ If $\eta(x)$ is a function that tends to infinity (that is, it is an infinite: $\eta(x) \rightarrow \pm\infty$), we have:

$$\left(1 + \frac{1}{\eta(x)}\right)^{\eta(x)} \rightarrow e$$

- ▶ It does not matter what x tends to

Example

Limits of functions tending to e

1.

$$\lim_{x \rightarrow +\infty} \left(1 + \frac{1}{2x^2 - 10} \right)^{2x^2 - 10} = e$$

since with $\eta(x) = 2x^2 - 10$ we have

$\eta(x) \rightarrow +\infty$ as $x \rightarrow +\infty$

2.

$$\lim_{x \rightarrow 0^+} \left(1 + \frac{1}{1/x} \right)^{1/x} = e$$

since with $\eta(x) = \frac{1}{x}$ we have $\eta(x) \rightarrow +\infty$ as $x \rightarrow 0^+$

From the fundamental limit

$$\lim_{x \rightarrow \pm\infty} \left(1 + \frac{1}{x}\right)^x = e$$

three more can be deduced: for the logarithm, for the exponential and for the power.

Fundamental limit of the logarithm

Corollary 1

$$\lim_{y \rightarrow 0} \frac{\log(1+y)}{y} = 1$$

Proof

Taking logarithms in $\lim_{x \rightarrow \pm\infty} \left(1 + \frac{1}{x}\right)^x = e$, we obtain

$$\log \left(1 + \frac{1}{x}\right)^x = x \log \left(1 + \frac{1}{x}\right)$$

hence

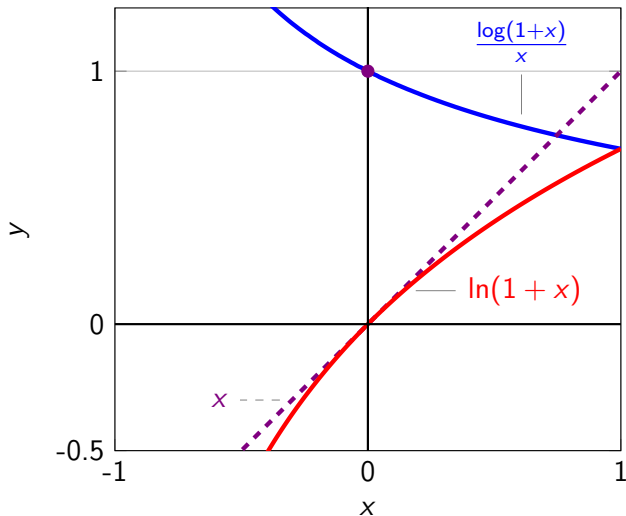
$$\lim_{x \rightarrow \pm\infty} x \log \left(1 + \frac{1}{x}\right) = \log e = 1.$$

Now, if we set $y = \frac{1}{x}$, $x \rightarrow \pm\infty$ is equivalent to $y \rightarrow 0^\pm$, and the last limit can therefore be rewritten in the following form:

$$\frac{\log(1+y)}{y} \rightarrow 1 \quad \text{as } y \rightarrow 0.$$



Graph of $\frac{\log(1+x)}{x}$



Fundamental limit of the exponential

Corollary 2

$$\lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1$$

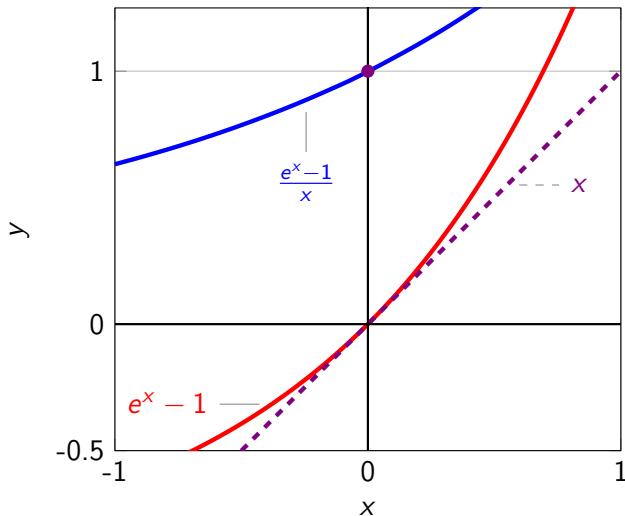
Proof

If in the fundamental limit of the logarithm $\lim_{y \rightarrow 0} \frac{\log(1+y)}{y} = 1$ we instead set $y = e^x - 1$, $y \rightarrow 0$ is equivalent to $x \rightarrow 0$, and substituting we obtain

$$\frac{\log e^x}{e^x - 1} = \frac{x}{e^x - 1} \rightarrow 1 \quad \text{as } x \rightarrow 0.$$

Taking reciprocals we obtain the fundamental limit. □

Graph of $\frac{e^x-1}{x}$



Example

Fundamental limit

We compute:

$$\lim_{x \rightarrow 0} \frac{e^{-x} - 1}{x}$$

We set $z = -x$, $x \rightarrow 0$ is equivalent to $z \rightarrow 0$, and substituting we obtain:

$$\lim_{x \rightarrow 0} \frac{e^{-x} - 1}{x} = \lim_{z \rightarrow 0} \frac{e^z - 1}{-z} = - \lim_{z \rightarrow 0} \underbrace{\frac{e^z - 1}{z}}_{\rightarrow 1} = -1$$

Fundamental limit of the power

Corollary 3

$$\lim_{x \rightarrow 0} \frac{(1+x)^\alpha - 1}{x} = \alpha \quad \text{with } \alpha \in \mathbb{R}$$

Proof (1/2)

If in the fundamental limit of the logarithm $\lim_{y \rightarrow 0} \frac{\log(1+y)}{y} = 1$ we instead set $y = (1+x)^\alpha - 1$, with α any real exponent, then $x \rightarrow 0$ is equivalent to $y \rightarrow 0$ and we have:

$$\begin{aligned} \frac{\log(1+y)}{y} &= \frac{\log[1 + (1+x)^\alpha - 1]}{(1+x)^\alpha - 1} = \frac{\alpha \log(1+x)}{(1+x)^\alpha - 1} \\ &= \frac{\alpha x}{(1+x)^\alpha - 1} \cdot \frac{\log(1+x)}{x} \rightarrow 1 \quad \text{as } x \rightarrow 0. \end{aligned}$$

Proof (2/2)

But since also

$$\frac{\log(1+x)}{x} \rightarrow 1 \quad \text{as } x \rightarrow 0,$$

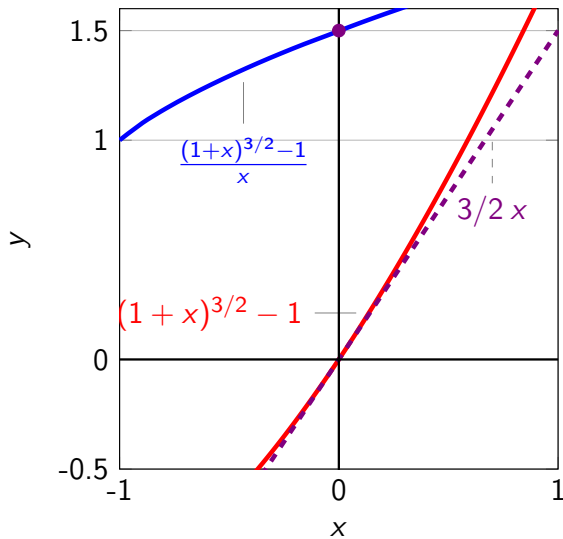
then

$$\frac{\alpha x}{(1+x)^\alpha - 1} \rightarrow 1 \quad \text{as } x \rightarrow 0.$$

Taking reciprocals and then multiplying by α we obtain the fundamental limit.



Graph of $\frac{(1+x)^{3/2}-1}{x}$



Summary (1/2)

Fundamental limits of sine and cosine:

- ▶ $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$ (proved by comparing areas and using the comparison theorem)
- ▶ $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} = \frac{1}{2}$ (deduced from the previous one)

Extension by continuity:

- ▶ If f is not defined at x_0 but has a finite limit ℓ as $x \rightarrow x_0$, setting $f(x_0) = \ell$ makes it continuous at x_0

Summary (2/2)

Fundamental limits related to the number e :

- ▶ $\lim_{x \rightarrow \pm\infty} \left(1 + \frac{1}{x}\right)^x = e$, and more generally $\left(1 + \frac{1}{\eta(x)}\right)^{\eta(x)} \rightarrow e$
if $\eta(x) \rightarrow \pm\infty$
- ▶ $\lim_{y \rightarrow 0} \frac{\log(1+y)}{y} = 1, \quad \lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1$
- ▶ $\lim_{x \rightarrow 0} \frac{(1+x)^\alpha - 1}{x} = \alpha$, with $\alpha \in \mathbb{R}$

Asymptotic estimates



Introduction: Asymptotic estimates

- ▶ In this section we introduce the concept of **asymptotic functions**: two functions whose ratio tends to 1
- ▶ The fundamental limits can be rewritten as **asymptotic estimates**, which tell us how certain elementary functions behave near 0
- ▶ Replacing the variable with any infinitesimal $\varepsilon(x)$, we obtain a very effective tool to **compute limits** in indeterminate form

Definition 1: of asymptotic functions

Two functions f, g are said to be **asymptotic** as $x \rightarrow c$ if

$$\lim_{x \rightarrow c} \frac{f(x)}{g(x)} = 1$$

and we write $f \sim g$ as $x \rightarrow c$.

- ▶ The asymptotic symbol for functions enjoys all the properties stated for sequences

Asymptotic estimates from the fundamental limits

As $x \rightarrow 0$ we have (from the fundamental limits):

$$\sin x \sim x$$

$$\log(1+x) \sim x$$

$$\cos x \sim 1 - \frac{1}{2}x^2$$

$$e^x \sim 1 + x$$

$$(1+x)^\alpha \sim 1 + \alpha x \quad \text{with } \alpha \in \mathbb{R}$$

- The functions $\sim x$ behave, to a first approximation or to first order, like x as $x \rightarrow 0$

Asymptotic estimates with an infinitesimal

If $\varepsilon(x)$ is a function that tends to zero (that is, it is an infinitesimal: $\varepsilon(x) \rightarrow 0$; it does not matter what x tends to), we have:

$$\sin(\varepsilon(x)) \sim \varepsilon(x)$$

$$\log(1 + \varepsilon(x)) \sim \varepsilon(x)$$

$$\cos(\varepsilon(x)) \sim 1 - \frac{1}{2} \varepsilon^2(x)$$

$$e^{\varepsilon(x)} \sim 1 + \varepsilon(x)$$

$$(1 + \varepsilon(x))^\alpha \sim 1 + \alpha \varepsilon(x) \quad \text{with } \alpha \in \mathbb{R}$$

- ▶ The estimates with $\varepsilon(x)$ follow from those as $x \rightarrow 0$ simply by a change of variable

$$x = \varepsilon(x)$$

- ▶ Let us now see how to use them to compute limits in indeterminate form

Example (1/2)

Limits with asymptotic functions

$$\lim_{x \rightarrow 1} \frac{(x-1)^2}{e^{3(x-1)^2} - 1} = \left[\frac{0}{0} \right]$$

We use the estimate

$$e^{\varepsilon(x)} - 1 \sim \varepsilon(x)$$

with

$$\varepsilon(x) = 3(x-1)^2 \quad \text{and} \quad \varepsilon(x) \rightarrow 0 \quad \text{as} \quad x \rightarrow 1$$

Example (2/2)

Then, as $x \rightarrow 1$, we have

$$e^{3(x-1)^2} - 1 \sim 3(x-1)^2$$

and

$$\lim_{x \rightarrow 1} \frac{(x-1)^2}{e^{3(x-1)^2} - 1} = \lim_{x \rightarrow 1} \frac{(x-1)^2}{3(x-1)^2} = \frac{1}{3}$$

Example (1/2)

Limits with asymptotic functions

$$\lim_{x \rightarrow 0} \frac{\log(1 + 2x)}{\sin 3x} = \left[\frac{0}{0} \right]$$

We use the estimate $\log(1 + \varepsilon(x)) \sim \varepsilon(x)$ with

$$\varepsilon(x) = 2x \quad \text{and} \quad \varepsilon(x) \rightarrow 0 \quad \text{as} \quad x \rightarrow 0.$$

Hence, as $x \rightarrow 0$ we have

$$\log(1 + 2x) \sim 2x$$

Example (2/2)

Now we also use the estimate $\sin \varepsilon(x) \sim \varepsilon(x)$ with

$$\varepsilon(x) = 3x \quad \text{and} \quad \varepsilon(x) \rightarrow 0 \quad \text{as} \quad x \rightarrow 0$$

Hence, as $x \rightarrow 0$ we have $\sin 3x \sim 3x$. Then, as $x \rightarrow 0$, we have

$$\frac{\log(1 + 2x)}{\sin 3x} \sim \frac{2x}{3x}$$

and

$$\lim_{x \rightarrow 0} \frac{\log(1 + 2x)}{\sin 3x} = \lim_{x \rightarrow 0} \frac{2x}{3x} = \frac{2}{3}$$

Example (1/2)

Limits with asymptotic functions

$$\lim_{x \rightarrow +\infty} \sqrt[3]{x^3 + 2x^2 + 1} - x = [+\infty - \infty]$$

We rewrite,

$$\sqrt[3]{x^3 + 2x^2 + 1} - x = x \left(\sqrt[3]{1 + \left(\frac{2}{x} + \frac{1}{x^3}\right)} - 1 \right)$$

We use the estimate

$$\sqrt[3]{1 + \varepsilon(x)} - 1 \sim \frac{1}{3}\varepsilon(x)$$

with

$$\varepsilon(x) = \left(\frac{2}{x} + \frac{1}{x^3}\right) \quad \text{and} \quad \varepsilon(x) \rightarrow 0 \quad \text{as} \quad x \rightarrow +\infty.$$

Example (2/2)

Then, as $x \rightarrow +\infty$, we have

$$\sqrt[3]{x^3 + 2x^2 + 1} - x \sim x \left[\frac{1}{3} \left(\frac{2}{x} + \frac{1}{x^3} \right) \right]$$

and

$$\lim_{x \rightarrow +\infty} \sqrt[3]{x^3 + 2x^2 + 1} - x = \lim_{x \rightarrow +\infty} x \left[\frac{1}{3} \left(\frac{2}{x} + \frac{1}{x^3} \right) \right] = \frac{2}{3}$$

Summary

Asymptotic functions:

► $f \sim g$ as $x \rightarrow c$ if $\lim_{x \rightarrow c} \frac{f(x)}{g(x)} = 1$

Asymptotic estimates:

- If $\varepsilon(x) \rightarrow 0$: $\sin \varepsilon(x) \sim \varepsilon(x)$, $\log(1 + \varepsilon(x)) \sim \varepsilon(x)$,
 $\cos \varepsilon(x) \sim 1 - \frac{1}{2}\varepsilon^2(x)$, $e^{\varepsilon(x)} \sim 1 + \varepsilon(x)$,
 $(1 + \varepsilon(x))^\alpha \sim 1 + \alpha \varepsilon(x)$
- To compute a limit in indeterminate form, we identify the infinitesimal $\varepsilon(x)$ and replace each factor with a simpler asymptotic function

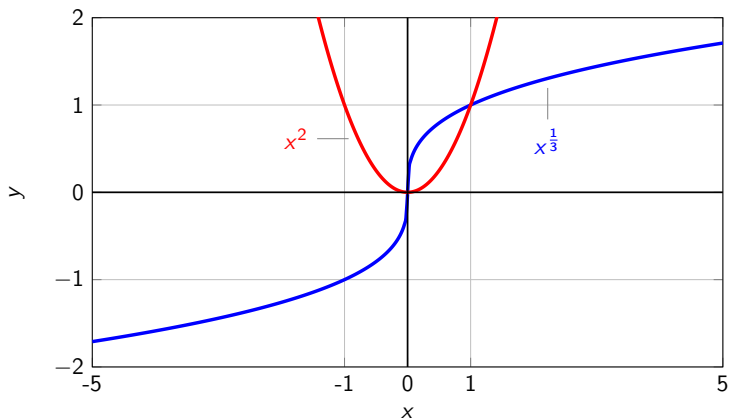
Asymptotic estimates and graphs

Introduction: Asymptotic estimates and graphs

- ▶ Asymptotic estimates are useful not only for computing limits, but also for sketching the **qualitative graph** of a function in a neighborhood of a given point, or as $x \rightarrow \pm\infty$
- ▶ We will compare the graph of a function with that of a known function to which it is asymptotic
- ▶ Finally, we will introduce the notions of **superlinear**, **linear** and **sublinear** growth at infinity

Example

Known graphs



Example (1/2)

Qualitative graph

We now want to study the qualitative graph of the function (the sum of the previous two):

$$f(x) = \sqrt[3]{x} + x^2 = x^{\frac{1}{3}} + x^2$$

- ▶ the function is defined and continuous on all of $\mathbb{R}$;
- ▶ as $x \rightarrow \pm\infty$, $f(x) \sim x^2$, since

$$\lim_{x \rightarrow \pm\infty} \frac{x^{\frac{1}{3}} + x^2}{x^2} = 1$$

Therefore $f(x) \rightarrow +\infty$ as $x \rightarrow \pm\infty$; moreover, for x large in absolute value, its graph will be similar to that of x^2 .

Example (2/2)

- Moreover, the function vanishes at $x = 0$ and, as $x \rightarrow 0$, $f(x) \sim x^{\frac{1}{3}}$, since

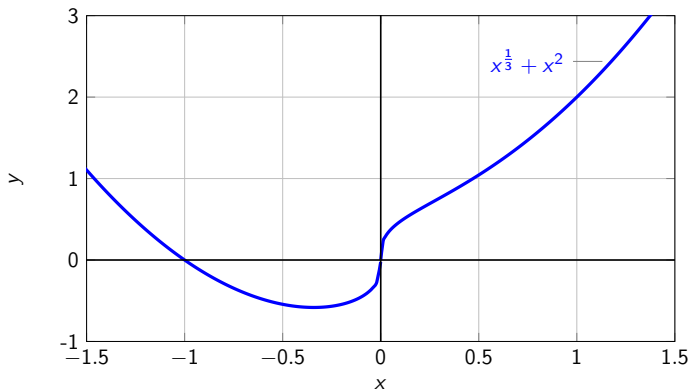
$$\lim_{x \rightarrow 0} \frac{x^{\frac{1}{3}} + x^2}{x^{\frac{1}{3}}} = 1$$

therefore, in a neighborhood of $x = 0$, its graph will be similar to that of $x^{\frac{1}{3}}$; in particular, it will have a vertical tangent at the origin.

Example

Actual graph

$$f(x) = \sqrt[3]{x} + x^2 = x^{\frac{1}{3}} + x^2$$



Asymptotic estimates and graphs

- ▶ Often the behavior of a function in a neighborhood of a point (for example, the fact that it has a vertical or horizontal tangent) can be predicted from a suitable asymptotic estimate
- ▶ The asymptotic estimate allows us to sketch the qualitative graph of f (in a neighborhood of the point) by comparison with that of a known function (for example, a power with rational exponent)
- ▶ Analogous estimates are useful as $x \rightarrow \pm\infty$

Growth of a function at infinity (1/2)

- ▶ Suppose we want to sketch the graph of a function f that, as $x \rightarrow +\infty$ (or $-\infty$), tends to $+\infty$ (or $-\infty$)
- ▶ To describe the speed at which the function tends to infinity, the following notions are useful: we say that, as $x \rightarrow +\infty$, the growth of f is

$$\left\{ \begin{array}{ll} \text{superlinear} & \text{if } \lim_{x \rightarrow +\infty} \frac{f(x)}{x} = \pm\infty \\ \text{linear} & \text{if } \lim_{x \rightarrow +\infty} \frac{f(x)}{x} = m \text{ (finite, } m \neq 0) \\ \text{sublinear} & \text{if } \lim_{x \rightarrow +\infty} \frac{f(x)}{x} = 0 \end{array} \right.$$

Growth of a function at infinity (2/2)

- ▶ Analogous definitions are given for $x \rightarrow -\infty$
- ▶ Only when a function has linear growth is it possible for it to have an oblique asymptote

Example

Growth of a function at infinity

For example, as $x \rightarrow +\infty$

- ▶ Superlinear growth: exponentials a^x and powers x^a with $a > 1$.
- ▶ Sublinear growth: logarithms $\log_a x$ and powers x^a with $0 < a < 1$.

Example

Growth of a function at infinity

The function

$$f(x) = 2x + e^x + e^{\frac{1}{x}}$$

- ▶ as $x \rightarrow +\infty$ it is asymptotic to e^x ; therefore it tends to $+\infty$ with superlinear growth;
- ▶ as $x \rightarrow -\infty$ it is asymptotic to $2x$; therefore it tends to $-\infty$ linearly
- ▶ since

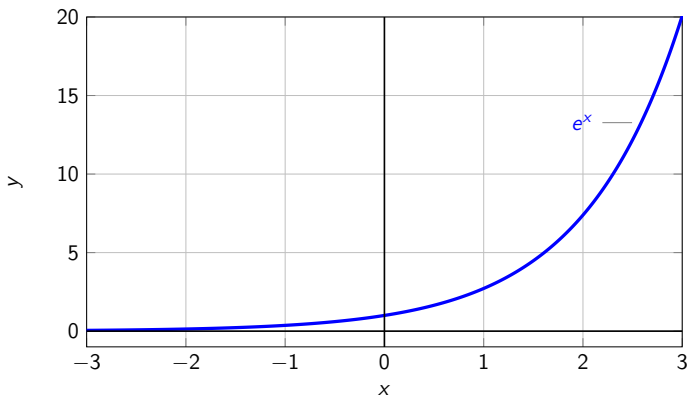
$$\lim_{x \rightarrow -\infty} [f(x) - 2x] = 1$$

the function has the oblique asymptote

$$y = 2x + 1 \quad \text{as } x \rightarrow -\infty$$

Example: graphs (1/2)

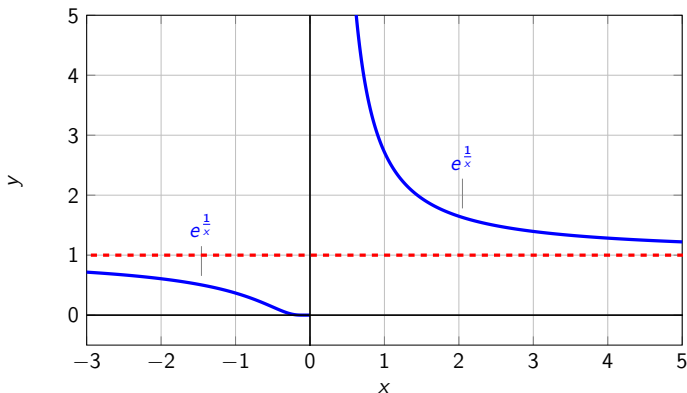
$$f(x) = e^x, \quad \lim_{x \rightarrow +\infty} e^x = +\infty, \quad \lim_{x \rightarrow -\infty} e^x = 0, \quad \lim_{x \rightarrow 0} e^x = 1$$



Example: graphs (2/2)

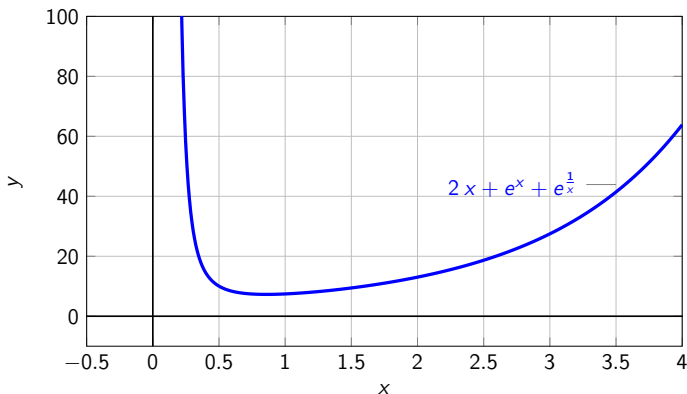
$$f(x) = e^{\frac{1}{x}}, \quad \lim_{x \rightarrow +\infty} e^{\frac{1}{x}} = 1, \quad \lim_{x \rightarrow -\infty} e^{\frac{1}{x}} = 1,$$

$$\lim_{x \rightarrow 0^-} e^{\frac{1}{x}} = 0, \quad \lim_{x \rightarrow 0^+} e^{\frac{1}{x}} = +\infty$$



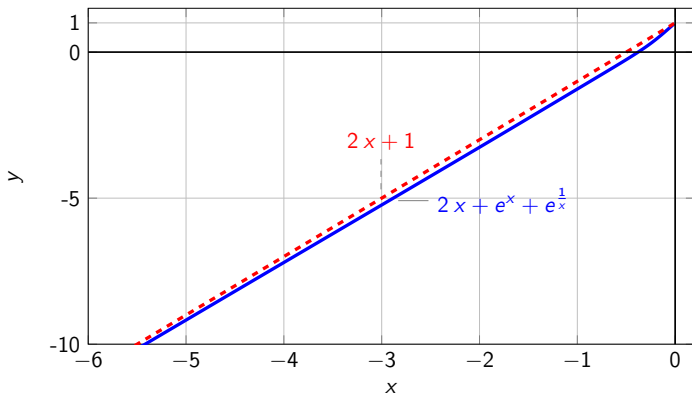
Example: actual graph (1/2)

$$f(x) = 2x + e^x + e^{\frac{1}{x}}, \quad f(x) \sim e^x \text{ as } x \rightarrow +\infty, \quad \lim_{x \rightarrow 0^+} f(x) = +\infty$$



Example: actual graph (2/2)

$$f(x) = 2x + e^x + e^{\frac{1}{x}}, \quad f(x) \sim 2x \text{ as } x \rightarrow -\infty, \quad \lim_{x \rightarrow 0^-} f(x) = 1$$



Summary

Asymptotic estimates and graphs:

- ▶ If $f \sim g$ near a point (or as $x \rightarrow \pm\infty$), the graph of f is similar to that of the known function g
- ▶ Example: $x^{\frac{1}{3}} + x^2 \sim x^2$ as $x \rightarrow \pm\infty$ and $\sim x^{\frac{1}{3}}$ as $x \rightarrow 0$ (vertical tangent at the origin)

Growth at infinity:

- ▶ Superlinear, linear or sublinear according to whether $\frac{f(x)}{x}$ tends to $\pm\infty$, to a finite $m \neq 0$, or to 0
- ▶ Only with linear growth can an oblique asymptote exist

Final summary

- ▶ **Fundamental limits:** $\frac{\sin x}{x} \rightarrow 1$, $\frac{1-\cos x}{x^2} \rightarrow \frac{1}{2}$, $\frac{\log(1+x)}{x} \rightarrow 1$, $\frac{e^x-1}{x} \rightarrow 1$, $\frac{(1+x)^\alpha-1}{x} \rightarrow \alpha$ as $x \rightarrow 0$, and $(1+\frac{1}{x})^x \rightarrow e$ as $x \rightarrow \pm\infty$
- ▶ **Extension by continuity:** possible only if the finite limit at the point exists
- ▶ **Asymptotic estimates:** the fundamental limits become $\sim$ relations valid for every infinitesimal $\varepsilon(x)$, useful to compute limits
- ▶ **Graphs:** asymptotic estimates allow us to sketch the qualitative graph and to classify the growth at infinity